

SAT UZBEKISTAN

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1.

For a survey, a sample of 1,018 people was selected at random from all registered voters in a county. Based on the survey, it is estimated that 66% of all registered voters in the county are in favor of a new law, with an associated margin of error of 2.91%. Which of the following best describes the plausible values for the percent of all registered voters in the county who are in favor of the new law?

(A) Any value greater than 63.09% and less than 68.91%

(B) Any value greater than 63.09%

(C) Any value less than 63.09% or greater than 68.91%

(D) Any value less than 68.91%

2.

The gas mileage of a certain car is 35 miles per gallon when driven on the highway. If the car begins with 10 gallons of gas in the tank, which function predicts the amount of gas remaining in the tank, $f(d)$, after the car is driven d miles on the highway?

(A) $f(d) = \frac{d}{35} - 10$

(B) $f(d) = d - 10$

(C) $f(d) = 10 - \frac{d}{35}$

(D) $f(d) = 10 - d$

3.

An equilateral triangle has a height of $47\sqrt{3}$ units. What is the length, in units, of one side of this triangle?

(A) $94\sqrt{3}$

(B) 94

(C) $47\sqrt{3}$

(D) 47

4.

$$f(x) = 2,000(1.005)^{3x}$$

The given function f models the balance of a bank account, in dollars, x years after it is opened. Which statement is the best interpretation of $(1.005)^{3x}$?

(A) Every 4 months, the balance increases by about \$5.

(B) At the end of every 4-month interval, the balance increases by about 0.5% of the balance at the beginning of the 4-month interval.

(C) Every 3 years, the balance increases by about \$5.

(D) At the end of every 3-year interval, the balance increases by about 0.5% of the balance at the beginning of the 3-year interval.

5.

A square map has a side length of 55 inches, and 1 inch on the map represents an actual distance of 13 miles. A larger version of the same map is printed as a square with the side length 90% longer than the side length of the previous map. On the larger map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

(A) 1.30

(B) 6.84

(C) 24.70

(D) 49.50

6.

$$\frac{4}{9}(4x+9)(x+\sqrt{4k+9})(x-\sqrt{4k+9})=0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 81. What is the value of k ?

7.

In isosceles triangle ABC, sides AB and AC are congruent, Point D divides side BC such that the length of $\overline{BD}$ is $\frac{4}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD. If the length of $\overline{BE}$ is 8, what is the length of $\overline{CF}$?

8.

$$x(kx - 112) = -16$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

9.

A researcher created a data set by recording the daily low temperature, to the nearest integer, for each of the 31 days in December 2022 in a town in Illinois. The frequency table summarizes this data set.

Daily low temperature (degrees Fahrenheit)	Frequency
−5 to 4	1
5 to 14	4
15 to 24	5
25 to 34	6
35 to 44	10
45 to 54	4
55 to 64	1

Based on the frequency table, what could be the median of the recorded daily low temperatures, in degrees Fahrenheit?

10.

The price of an object increased from **\$16** to **\$80**. What was the percent increase in the price of the object?

11.

A real estate company offers a series of three webinars. **1,250** people attended the first webinar. **46%** of the people who attended the first webinar attended the second webinar, and **32%** of the people who attended the first and second webinars attended the third webinar. How many people attended all three webinars?

12.

In the xy -plane, line k and line l are perpendicular and intersect at the point $(2, 8)$. If line k is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line l ?

☐ (A) $\left(3, 8 - \frac{1}{m}\right)$

☐ (B) $\left(3, 8 + \frac{1}{m}\right)$

☐ (C) $(3, 8 - m)$

☐ (D) $(3, 8 + m)$

13.

$$f(x) = 7020(0.28)^{\frac{x}{12}}$$

The function f gives the value of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p ?

A) 2

C) 28

B) 10

D) 72

14.

The function f and g are defined by the equations, where $x > 0$. Which of the following equations displays, as a constant or a coefficient, the maximum value of the function it defines, where $x \geq 0$?

I. $f(x) = 16(1.25)^x + 43$

II. $g(x) = 6(0.75)^x$

- A) I only
B) II only
C) I and II
D) Neither I nor II

15.

The equation $N(m) = 65(Q)^{\frac{m}{4}}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 120 seconds. What is the value of p in terms of Q ?

- A) $100(Q^{\frac{1}{2}} + 1)$
B) $100(Q^{30} + 1)$
C) $100(Q^{\frac{1}{2}} - 1)$
D) $100(Q^{30} - 1)$

16.

$$(x - k)^2 = (k - 4a)(x - k)$$

In the given equation, a and k are constants, where $k > 4a$. The sum of the solutions to the equation is $3k + 35$. What is the value of a ?

17.

$$P(t) = 260(1.03)^{\left(\frac{6}{n}\right)t}$$

The function P models the population, in thousands, of a certain city t years after 2006. According to the model, the population is predicted to increase by 3% every n months. What is the value of n ?

- A) 8
B) 12
C) 18
D) 72

18.

One map of a town is drawn to the scale 8 centimeters (cm) = 1 kilometer (km). A second map of the town is drawn to the scale 24 cm = 1 km. On the first map, the length of a street measures x cm. Which expression represents the length of this street, in centimeters, on the second map?

One map

(A) $\frac{1}{3}x$

(B) $3x$

(C) $8x$

(D) $192x$

19.

$f(x) = 3(x - a)(x - b)(x - c)$

The function f is defined by the given equation, where a , b , and c are distinct constants. When $a < x < b$, the value of $f(x)$ is positive. The graph of $y = f(x)$ in the xy -plane contains the point (r, s) , where r and s are constants. If $s = 8$, which of the following could be true?

I. $r < a$
 II. $b < r < c$
 III. $r > c$

(A) I only

(B) III only

(C) I and III only

(D) II and III only

20.

A right square pyramid has a total surface area of 36,864 square inches, and the combined surface area of the four lateral faces of this pyramid is 20,480 square inches. What is the height, in inches, of this pyramid?

21.

The perimeter of an equilateral triangle is 642 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

22.

$$8rx^2 - 32sx + 8r = 0$$

For the given equation, r and s are constants such that $r > 0$, $s < 0$, and $\frac{r}{s} > k$. What is the least value of k that ensures the equation has two distinct real solutions?